Shaon Mandal Chakraborty

PhD Student · Institut für Theoretische Physik II - Soft Matter

Heinrich-Heine-Universität, Düsseldorf

Shaon.Chakraborty@hhu.de

Education

Heinrich Heine University Düsseldorf *Düsseldorf, Germany*
PhD in Physics *August, 2026 - present*

- Advisor: Prof. Jürgen Horbach

University of Calcutta *Calcutta, India*
Masters in Physics *July, 2021 - June, 2023*

- CGPA: 8.33
- Thesis title: Branching of Trajectories in a Hopfield-like Model
- Advisor: Dr. Archishman Raju

University of Calcutta *Calcutta, India*
BSc (Undergraduate Degree) in Physics *July, 2018 - May, 2021*

- Minors in Mathematics and Chemistry
- CGPA: 7.75
- Thesis title: The Effect of Dark Matter – Baryon Interaction on Primordial Black-holes
- Advisor: Prof. Debasish Majumdar

Employment

Raman Research Institute *Bangalore, India*
Research Assistant *March, 2025 - July, 2026*
Advisor: Dr. Rituparno Mandal

- Working on the effects of non-reciprocity in disordered systems.

Raman Research Institute *Bangalore, India*
Visiting Student *November, 2024 - March, 2025*
Advisor: Dr. Rituparno Mandal

- Studied and explored a model of physical learning in an interacting spin system.
- Analyzed the effects of non-reciprocal coupling in Kuramoto oscillators, focusing on the emergence and stability of chimera states.

National Centre for Biological Sciences, TIFR *Bangalore, India*
Research Intern *July 2023 – August 2024*
Advisor: Dr. Archishman Raju

- Analyzed mitochondrial network morphology through the lens of percolation theory, renormalization group techniques, and critical phenomena associated with phase transitions.

National Centre for Biological Sciences, TIFR *Bangalore, India*
Master's Research Intern *Jan 2023 – May 2023*
Advisor: Dr. Archishman Raju

- Developed an extended Hopfield-type model capable of storing dynamical trajectories linking multiple fixed-point attractors.

Publications & Preprints

- **Shaon Mandal Chakraborty**, Bibhut Sahoo, Peter Sollich, Rituparno Mandal,
  Effects of Non-reciprocity on Coupled Kuramoto Oscillators,
  arXiv:2511.15845 (2025).

Skills

- **Languages:** English, Bengali, Hindi
- **Programming & Tools:** Python, R, Julia, Fortran, Mathematica, LAMMPS, SQL, LaTeX
- **Scientific Computing:** Data analysis and visualization, high-performance computing (SLURM), MPI, OpenMP
- **Relevant Coursework:** Advanced Statistical Mechanics, Soft Condensed Matter, Advanced Quantum Mechanics, Advanced Classical Mechanics, Mathematical Methods in Physics, Electrodynamics, Numerical Methods, Materials Science, Quantum Condensed Matter, Theoretical Ecology

Co-curricular Experience

Teaching Experience

Teaching Assistant
Theoretical Soft Condensed Matter (Fall, 2025)
Raman Research Institute
Course Instructor: Dr. Rituparno Mandal

Certification

Summer School in Theoretical Physics
Indian Association for the Cultivation of Science
2018

References

Dr. Rituparno Mandal
Associate Professor
Department of Soft Condensed Matter Physics
Raman Research Institute
Bangalore
`rituparno@rri.res.in`

Prof. Jürgen Horbach
Professor
Institut für Theoretische Physik II - Soft Matter
Heinrich Heine University Düsseldorf
Düsseldorf
`horbach@thphy.uni-duesseldorf.de`